CHAPTER 9

Detection and Measurement Techniques

Contents

9.1. Track Measurements 240
9.1.1. Cloud and bubble chambers 240
9.1.2. Solid state nuclear track detectors (SSNTD) 241
9.2. General Properties of Detectors 246
9.2.1. Pulse generation 247
9.2.2. Basic counting systems 249
9.2.3. Pulse shape and dead time 250
9.3. Gas Counters 254
9.3.1. Ion chambers 255
9.3.2. Proportional counters 258
9.3.3. Geiger-Müller counters 260
9.4. Semiconductor Detectors 262
9.4.1. Surface barrier detectors 264
9.4.2. Lithium-drifted detectors 266
9.4.3. Intrinsic detectors 268
9.5. Scintillation Detectors 268
9.5.1. Gas scintillator detectors 271
9.5.2. Liquid scintillator detectors 272
9.5.3. Solid scintillator detectors 273
9.6. Čerenkov Detectors 274
9.7. Microchannel Plate Detectors 274
9.8. Electronics for Pulse Counting 275
9.8.1. Preamplifiers 275
9.8.2. Amplifiers 276
9.8.3. Single channel analyzers 277
9.8.4. Counters and rate meters 277
9.8.5. Multichannel analyzers 278
9.8.6. γ-spectrometry 279
9.9. Special Counting Systems 283
9.10. Absolute Disintegration Rates 285
9.11. Sample Preparation 286
9.12. Statistics of Counting and Associated Error 288
9.12.1. One-atom-at-a-time measurements 291
9.13. Exercises 293
9.14. Literature 295

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The ionization and/or excitation of atoms and molecules when the energies of nuclear particles are absorbed in matter is the basis for the detection of individual particles. Macroscopic collective effects, such as chemical changes and heat evolution, can also be used. The most important of the latter have been described before because of their importance for dose measurements (e.g. the blackening of photographic films and other chemical reactions, excitation of crystals (thermoluminescence), and heat evolved in calorimeters; Ch. 8). Although animals have no known senses for detection of nuclear radiation, it has been found that subletal but large radiation fields can affect animals in various ways such as disturbing the sleep of dogs or causing ants to follow a new pathway to avoid a hidden radiation source. Apollo astronauts observed scintillations in their eyes when their space ship crossed very intense showers of high energy cosmic rays. People who have been involved in criticality accidents experiencing high intensities of n and γ have noted a fluorescence in their eyes and felt a heat shock in their body.

However, we are not physiologically aware of the normal radiation fields of our environment. In such low fields we must entirely rely on instruments.

In this chapter we consider only the common techniques used for detection and quantitative measurement of *individual* nuclear particles. We also discuss the problem of proper preparation of the sample to be measured as well as consideration of the statistics of the counting of nuclear particles necessary to ensure proper *precision* (i.e. how well a value is determined) and *accuracy* (i.e. agreement between measured and true value).

9.1. TRACK MEASUREMENTS

The most striking evidence for the existence of atoms comes from the observation of tracks formed by nuclear particles in cloud chambers, in solids and in photographic emulsions. The tracks reveal individual nuclear reactions and radioactive decay processes. From a detailed study of such tracks, the mass, charge and energy of the particle can be determined.

The tracks formed can be directly observed by the naked eye in cloud and bubble chambers, but the tracks remain only for a short time before they fade. For a permanent record we must use photography. On the other hand, in solid state nuclear track detectors (SSNTD), of which the photographic emulsion is the most common variant, the tracks have a much longer lifetime during which they can be made permanent and visible by a suitable chemical treatment. Because of the much higher density of the absorber, the tracks are also much shorter and often therefore not visible for the naked eye. Thus the microscope is an essential tool for studying tracks in solids.

9.1.1. Cloud and bubble chambers

The principle of a *cloud chamber* is shown in Figure 9.1. A volume of saturated vapour contained in a vessel is made supersaturated through a sudden adiabatic expansion. When ionizing radiation passes through such a supersaturated vapour the ionization produced